

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)**

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1003	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Test Case Design Assignment

Topic: DevOps-Based Automated Competitive Exams Preparation and Learning Management Platform

Question 1: Identify the Functional and Non-Functional Requirements to be Tested.

The **DevOps-Based Automated Competitive Exams Preparation and Learning Management Platform** is designed to help students prepare for competitive examinations through online learning materials, practice tests, automated evaluation, performance analysis and progress tracking. The following functional and non-functional requirements need to be tested.

Functional Requirements

1. **User Registration:** The system should allow new users to create an account using valid personal details, email ID and password.
2. **User Login:** Registered users should be able to log in using valid credentials.
3. **Learning Materials:** Users should be able to access courses, study materials, videos, notes and practice resources.
4. **Online Examination:** Users should be able to select and attend competitive examinations.
5. **Automatic Evaluation:** The system should automatically evaluate submitted answers.
6. **Score Calculation:** The system should calculate and display the user's marks, percentage and result.
7. **Performance Tracking:** Users should be able to view their performance and learning progress.
8. **Question Management:** Administrators should be able to add, update and delete examination questions.
9. **Exam Management:** Administrators should be able to create and manage competitive examinations.
10. **Result Management:** The system should securely store examination results.
11. **Notification:** The system should notify users about examinations, results and learning activities.

12. **DevOps Automation:** The system should support automated build, testing and deployment through a CI/CD pipeline.

Non-Functional Requirements

1. **Performance:** The system should provide responses within an acceptable time.
2. **Security:** User accounts, passwords and examination results should be protected from unauthorized access.
3. **Reliability:** The system should work correctly without frequent failures.
4. **Availability:** The platform should be available whenever users need to attend examinations.
5. **Scalability:** The system should support an increasing number of students and examinations.
6. **Usability:** The platform should have a simple and user-friendly interface.
7. **Maintainability:** The software should be easy to modify and maintain.
8. **Recoverability:** The system should recover properly from server, network or deployment failures.

Question 2: Identify the Test Scenarios for the Assigned Problem Statement

A **test scenario** describes a functionality or situation that needs to be tested in the system. The major test scenarios for the DevOps-Based Automated Competitive Exams Preparation and Learning Management Platform are given below.

1. Verify that a new user can register successfully.
2. Verify that a registered user can log in using valid credentials.
3. Verify that the system rejects invalid login credentials.
4. Verify that the user can log out successfully.
5. Verify that users can access learning materials.
6. Verify that users can select and start competitive examinations.
7. Verify that users can answer examination questions.
8. Verify that unanswered questions are handled correctly.
9. Verify that the examination timer works correctly.
10. Verify that the examination is automatically submitted when the timer expires.
11. Verify that answers are automatically evaluated.

12. Verify that marks and percentage are calculated correctly.
13. Verify that examination results are displayed correctly.
14. Verify that user learning progress is updated.
15. Verify that administrators can add examination questions.

Question 3: Prepare Detailed Test Cases Covering Normal, Boundary, Invalid and Exceptional Conditions.

Test cases are prepared to verify whether the system behaves as expected under different conditions. The following test cases cover normal, boundary, invalid and exceptional conditions.

Test Case 1 – User Registration

Test Case ID: TC001

Test Scenario: Verify successful user registration.

Test Steps:

1. Open the platform.
2. Click on the Registration option.
3. Enter the user's name.
4. Enter a valid email address.
5. Enter a valid password.
6. Click the Register button.

Test Data:

Name: Ravi

Email: ravi@gmail.com

Password: Ravi@123

Expected Result:

The system should successfully create the user account and display a registration success message.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 2 – Invalid Email Registration

Test Case ID: TC002

Test Scenario: Verify registration with an invalid email address.

Test Steps:

1. Open the registration page.
2. Enter valid user details.
3. Enter an invalid email address.
4. Click Register.

Test Data: ravi@gmail

Expected Result:

The system should reject the email address and display an appropriate validation message.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 3 – Duplicate Email Registration

Test Case ID: TC003

Test Scenario: Verify registration using an already registered email.

Test Steps:

1. Open the registration page.
2. Enter an existing email address.
3. Enter other valid details.
4. Click Register.

Test Data: existing@gmail.com

Expected Result:

The system should not create another account and should display a duplicate email error.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 4 – Valid Login

Test Case ID: TC004

Test Scenario: Verify login using valid credentials.

Test Steps:

1. Open the login page.

2. Enter a registered email.
3. Enter the correct password.
4. Click Login.

Test Data:

Email: ravi@gmail.com

Password: Ravi@123

Expected Result:

The user should be successfully logged in and redirected to the dashboard.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 5 – Invalid Login

Test Case ID: TC005

Test Scenario: Verify login using an incorrect password.

Test Steps:

1. Open the login page.
2. Enter a registered email.
3. Enter an incorrect password.
4. Click Login.

Test Data:

Email: ravi@gmail.com

Password: Wrong@123

Expected Result:

The system should reject the login and display an invalid credentials message.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 6 – Access Learning Material

Test Case ID: TC006

Test Scenario: Verify that a registered user can access learning materials.

Test Steps:

1. Login to the platform.

2. Open the Learning section.
3. Select a competitive examination course.
4. Select study material.

Test Data: Quantitative Aptitude – Study Material.

Expected Result:

The selected learning material should open successfully.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 7 – Start Competitive Exam

Test Case ID: TC007

Test Scenario: Verify that a user can start an examination.

Test Steps:

1. Login to the platform.
2. Select Competitive Exams.
3. Select an available exam.
4. Click Start Exam.

Test Data: Quantitative Aptitude Test.

Expected Result:

The examination should start and the questions should be displayed.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 8 – Correct Answer Evaluation

Test Case ID: TC008

Test Scenario: Verify automatic evaluation of a correct answer.

Test Steps:

1. Start an examination.
2. Read the question.

3. Select the correct answer.
4. Submit the answer.

Test Data: Correct option selected.

Expected Result:

The system should record the answer and award the appropriate marks.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 9 – Wrong Answer Evaluation

Test Case ID: TC009

Test Scenario: Verify evaluation of an incorrect answer.

Test Steps:

1. Start an examination.
2. Select an incorrect option.
3. Submit the answer.

Test Data: Incorrect option.

Expected Result:

The system should record the answer and calculate marks according to the examination rules.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 10 – Unanswered Question

Test Case ID: TC010

Test Scenario: Verify handling of unanswered questions.

Test Steps:

1. Start an examination.
2. Leave a question unanswered.
3. Complete the examination.
4. Submit the test.

Test Data: No option selected.

Expected Result:

The system should treat the question as unanswered and calculate the result according to the examination rules.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 11 – Examination Timer

Test Case ID: TC011

Test Scenario: Verify examination timer functionality.

Test Steps:

1. Start a timed examination.
2. Observe the timer.
3. Continue the examination until the timer reaches zero.

Test Data: 30-minute examination.

Expected Result:

The timer should count down correctly. When the timer reaches zero, the examination should be automatically submitted.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 12 – Score Calculation

Test Case ID: TC012

Test Scenario: Verify automatic score calculation.

Test Steps:

1. Attend an examination containing 20 questions.
2. Answer 15 questions correctly.
3. Submit the examination.
4. View the result.

Test Data: 20 questions, 15 correct answers.

Expected Result:

The system should calculate and display the correct score and percentage according to the examination marking scheme.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Test Case 13 – Performance Report

Test Case ID: TC013

Test Scenario: Verify performance report generation.

Test Steps:

1. Complete an examination.
2. Submit the examination.
3. Open the Performance Report section.

Test Data: Completed examination result.

Expected Result:

The system should display the user's score, percentage and relevant performance information.

Actual Result: To be filled during testing.

Status: Pass/Fail to be recorded after execution.

Question 4: Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result and Pass/Fail Status.

Answer:

Each test case in the proposed platform contains the following important fields:

- **Test Case ID:** A unique identification number for each test case.
- **Test Scenario:** Describes the functionality or condition being tested.
- **Test Steps:** Provides the sequence of actions required to execute the test.
- **Test Data:** Contains the input values required for testing.
- **Expected Result:** Describes the result that should be obtained.
- **Actual Result:** Records the result obtained during actual execution.

- **Pass/Fail Status:** Indicates whether the test case passed or failed.

The actual result and status can be filled during the practical execution of the test cases.

Question 5: Apply Appropriate Test Design Techniques.

Answer:

Different test design techniques are applied to provide better coverage of the platform.

5.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups.

For example, if the platform allows a password length between **8 and 20 characters**, the input classes are:

Input Class	Example	Expected Result
Less than 8	Abc@12	Invalid
8–20 characters	Abc@1234	Valid
More than 20	21+ characters	Invalid

This technique reduces the number of test cases while maintaining effective input coverage.

5.2 Boundary Value Analysis

Boundary Value Analysis tests values at the limits of an input range.

For a **30-minute examination**, important boundary conditions are:

Test Value	Expected Result
29 minutes	Examination continues
30 minutes	Valid examination duration
30:00 timer expiry	Examination is submitted

More than allowed duration Should not be permitted

Boundary testing is especially useful for examination duration, number of questions, marks and input limits.

5.3 Decision Table Testing

Decision Table Testing is used when the system's output depends on multiple conditions.

Example: Result Generation

Condition	Rule 1	Rule 2	Rule 3
Examination submitted	Yes	Yes	No
Passing score achieved	Yes	No	-
Result generated	Yes	Yes	No
Display Pass	Yes	No	No
Display Fail	No	Yes	No

This ensures that different combinations of examination conditions are tested.

5.4 State Transition Testing

State Transition Testing verifies how the system changes from one state to another.

Examination States:

Not Started → Started → Answering → Submitted → Evaluated → Result Generated

For example:

- When the user clicks **Start Exam**, the state changes from Not Started to Started.
- When the user answers questions, the state remains Answering.
- When the user clicks Submit, the state changes to Submitted.
- The system then evaluates the answers.
- Finally, the result is generated.

This technique ensures that incorrect state transitions are identified.

Question 6: Ensure Adequate Requirement and Functional Coverage.

Requirement and functional coverage ensures that all important functions and requirements of the platform are tested at least once. The proposed test cases cover

user registration, login, learning materials, competitive examinations, automatic evaluation, score calculation, performance tracking, administration, security, notifications and DevOps-based automated deployment.

Requirement	Test Cases	Coverage
User Registration	TC001–TC003	Covered
User Login	TC004–TC005	Covered
Learning Materials	TC006	Covered
Competitive Examination	TC007–TC011	Covered
Automatic Evaluation	TC008–TC012	Covered
Performance Report	TC013	Covered
Question Management	TC014	Covered
Security	TC015	Covered
Failure Recovery	TC016	Covered
Automated Deployment	TC017	Covered
Failed Deployment	TC018	Covered

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(1 5%)	Clearly identifies all relevant functional/n	Identifies most requireme	Identifies basic requirements	Requirements/scenarios are incomplete or unclear

	on-functional requirements and comprehensive test scenarios	nts and scenarios	and scenarios	
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied

Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25